

PHYS 661 Fall 2024

Problem set 4, due October 17 2024. Each problem is 10 pts unless otherwise stated. Submit your homework either (1) via Brightspace or by (2) emailing it directly to the TA at gbowling@purdue.edu.

1 Harmonic oscillator wave functions in the WKB approximation

In the lecture we calculated the energy of the 1D harmonic oscillator [potential $V(x) = \frac{1}{2}m\omega_0^2x^2$] in the WKB approximation. Compare numerically the WKB wave function $\psi_n^{(\text{WKB})}(0)$ at the origin to the exact harmonic oscillator wave function for $n = 0, 1, 2, \dots, 8$. Interpret your results. The WKB wave function is normalized as follows (no need to prove that this is the correct normalization): in the classically allowed region $\psi^{(\text{WKB})}(x) = \frac{A}{\sqrt{p(x)}} \cos(\frac{1}{\hbar} \int_b^x dx' p(x') - \frac{\pi}{4})$ has $A^2 = [\int_b^a \frac{1}{2p(x)} dx]^{-1}$ where a, b are the turning points.

2 Number of bound states of a potential

Consider the 1D potential $V(x) = V_0 \text{sech}^2(x/a)$ with $V_0 < 0$.

1. Find the bound state energies in the WKB approximation. What is the number of bound states as a function of V_0 ?
2. For a generic 1D potential $U(x)$, by using the WKB approximation, estimate the number of bound states. Does your result agree with the estimate you get from part 1 above? How about in the case of the harmonic potential?

3 Vibration of NH_3 (5 pts)

The ammonia molecule NH_3 forms a shallow pyramid structure with a triangular base of hydrogen and nitrogen on the top. The nitrogen on top vibrates like a harmonic oscillator with frequency 950 cm^{-1} [in wavenumber units] separating the two lowest modes. In practice, the lowest two modes are seen to have small splittings of 0.8 cm^{-1} and 36 cm^{-1} , respectively. Interpret the splitting to be arising from N tunneling from the top of the pyramid to the bottom (inverting the pyramid upside-down), estimate the value for $\phi = \frac{1}{\hbar} \int_{-x_i}^{x_i} dx' |p(x')|$ for the two modes. Comment on the result.